



CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

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I. Record of Changes

Date	A* M, D	In charge	Change Description
24/07/2024	A	Ngo Anh Duy	Create report item(1,2,3,4,5)

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

Table 1.1 : Features and Functions to be tested

Feature	Function
User and Access Management	Role and Permission management
	Authentication and authorization
Schedule Management	Schedule an appointment online
	Doctor's work schedule Management
Medical records management	Store medical record results
	Search medical record results
Online consulting	Online medical examination
	Reminder for a follow-up appointment
Marketing	Customer feedback and question management
	Medical information blog management

Table 1.2: Test stage

Test Stage	In-Charge	Inputs/Time	Acceptance criteria
Unit test	Tester	Individual code units or modules	All unit test cases must be tested and passed 100% to pass this stage. All defects should be fixed and re-tested. Average of 11 bugs/KLOC.
Integration test	Tester	Integrated modules or components	All test cases must be tested and passed 100% to pass this stage. All defects should be fixed and re-tested. Average of 4 bugs/KLOC.
System test	Tester	Complete Pharmacy Distribution System	All test cases must be tested and passed 100% to pass this stage. All defects should be fixed and re-tested. Average of 0.5 bugs/KLOC

2. Test Strategy

2.1 Testing Types

2.1.1. API Test

Table 2.1.1.1 Api test

Test Objective:	This testing aims to ensure that the APIs correctly handle requests and responses, effectively communicate with external systems, and maintain data integrity.
Technique:	- API testing will be conducted using a combination of manual and automated testing approaches. - Automated testing tools, such as Postman and Swagger, will be employed to execute test scenarios, simulate various API requests and responses, and validate the expected outcomes. - Manual testing will also be performed to assess more complex scenarios and to evaluate the human experience of using the APIs.
Completion Criteria:	-All API endpoints have been tested, and test cases cover a wide range of scenarios including positive, negative, boundary, and error cases. -The API documentation is accurate and up-to-date, reflecting the correct usage and behavior of each API endpoint
Special Considerations:	N/A

2.1.2. User interface test

Table 2.1.2.1 User interface test

Test Objective:	Navigation within the subject under assessment accurately mirrors business operations and specifications, encompassing transitions between screens, fields, and utilization of entry points. Elements and attributes like menus, dimensions, arrangement, condition, and emphasis align with guidelines governing user interface and design that the customer has approved
Technique:	- Compare the current interface with a previous version or with similar interfaces in the industry to identify areas for improvement. - Evaluate the graphical user interface (GUI) elements, such as buttons, checkboxes, radio buttons, and dropdowns, to ensure they behave correctly. - Verify the flow of navigation between different screens, menus, and features to ensure a smooth user journey.

Completion Criteria:	- Elements, such as menus, buttons, text fields, and graphics, are positioned and styled correctly according to UI/UX design that the customer has approved. - Users can seamlessly move between screens, menus, and features without encountering any unexpected behaviors or broken links - The language of the information displayed to the user is Vietnamese
Special Considerations:	N/A

2.1.3. Functional Test

Table 2.1.3.1 Functional Test

Test Objective:	This type of testing is aimed at verifying that all the components, modules and interactions in the system work seamlessly with each other and meet the specified requirements.
Technique:	- The tester will use the black box testing technique - Test cases will be created based on the function description in SRS
Completion Criteria:	The entire software system functions as expected and meets the specified requirements.
Special Considerations:	N/A

2.1.4. Business Process Test

Table 2.1.4.1 Business Process Test

Test Objective:	Verify the application with business processes to ensure the system functions meet Customer's requirements
Technique:	- Testers will create test scenarios against the requirements provided by the workflow. Test scenarios will be created based on black box test technique. - Processes need to function properly
Completion Criteria:	Validate that the entire business process, from initiation to completion, functions correctly and accurately within the software. Ensure that the software correctly follows the defined business rules, validations, and decision points throughout the process.

Special Considerations:	N/A
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2.2 Test Levels

Table 2.2.1 Test levels

Type of Tests	Test Level			
	Unit	Integration	System	Acceptance
API Test	X			
User interface test		X		
Functional Test		X		
Business Process Test			X	

2.3 Supporting Tools

Table 2.3.1 Supporting tools

Purpose	Tool	Vendor/In-house	Version
Document(Manage test case and test report)	Google Sheet	Google	N/A
API test	Postman	Postman Inc	10.12.0
Functional, Business process and UI test	Google Chrome	Google	116.0.5845.96

3. Test Plan

3.1 Human Resources

Table 3.1.1 Human resources

Worker/Doer	Role	Specific Responsibilities/Comments
Ngo Anh Duy	Test Leader	Defining goals, determining workload, allocating resources Test planning, define procedures for test plan Review and evaluate the tasks of the test process Analysis and design of Test Cases Prepare for test Data Execute test

		Test Recording
Nguyen Duc Phuong	Tester	Defining goals, determining workload, allocating resources Test planning, define procedures for test plan Review and evaluate the tasks of the test process Analysis and design of Test Cases Prepare for test Data Execute test Test Recording
Nguyen An Thuyen	Tester	Defining goals, determining workload, allocating resources Test planning, define procedures for test plan Review and evaluate the tasks of the test process Analysis and design of Test Cases Prepare for test Data Execute test Test Recording

3.2 Test Environment

Table 3.2.1 Test environment

Name	Hardware/Software	Provider	Version
Laptop Dell G7 7588	Hardware	Nguyen Duc Phuong	window 10,8Gb RAM, Core i7-8750H
Laptop Gigabyte GE G5	Hardware	Ngo Anh Duy	window 11, 16Gb RAM , Core i5-12500H
ASUS TUF fx505dt	Hardware	Nguyen An Thuyen	window 11, 16Gb RAM, AMD Ryzen 7

3.3 Test Milestones

Table 3.3.1 Test milestones

Milestone Task	Start Date	End Date
Unit test	24/07/2024	29/07/2024
Integration test	07/08/2024	08/08/2024
System test	09/08/2024	09/08/2024

4. Test Cases

Unit Test:

<https://docs.google.com/spreadsheets/d/13bGu0zLdz0RM8Y4xFZtEcVgBF8SJIO0D/edit?gid=736026481#gid=736026481>

Integration Test:

https://docs.google.com/spreadsheets/d/1Auu_WYDvdR15cpj15gYMRS1EioVwlg89/edit?gid=1309820641#gid=1309820641

System Test:

https://docs.google.com/spreadsheets/d/1Eel2xhOoKkTq946_r1rfqDCis9f9WzzC/edit?gid=180337843#gid=180337843

5. Test Reports

5.1 Unit test

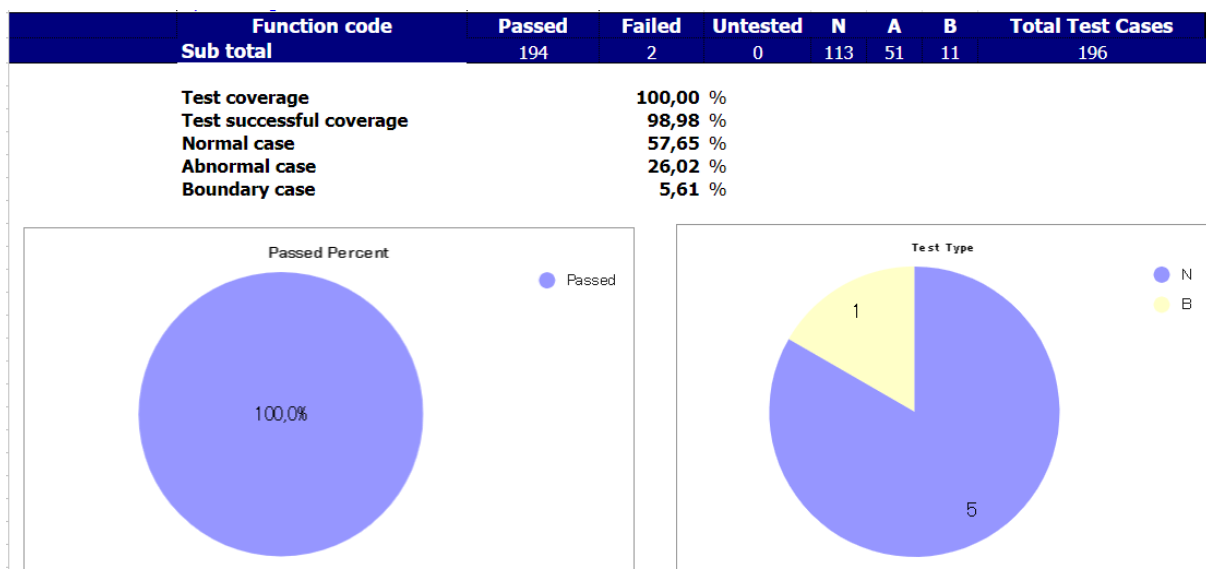


Figure 5.1.1 Unit test

5.2 Integration test

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Schedule Management	50	0	0	0	50
2	Online Consulting	20	0	0	0	20
3	Medical Notebooks man	20	0	0	0	20
4	Marketing	27	0	0	0	27
5	User and Access Manag	51	0	0	0	51
Sub total		168	0	0	0	168

Test coverage	100,00 %
Test successful coverage	100,00 %

Figure 5.2.1 Integration test

5.3 System test

No	Module code	Passed	Failed	Pending	N/A	Number of test cases
1	Appointment	12	0	0	0	12
2	Online Advising	7	0	0	0	7
3	StoreMedicalNotebook	7	0	0	0	7
4	DoctorSchedule	4	0	0	0	4
5	Ask doctor question	8	0	0	0	8
6	Post treatment	10	0	0	0	10
Sub total		26	0	0	0	26

Test coverage 100,00 %
Test successful coverage 100,00 %

Figure 5.3.1 System test